

PHARMACOEPIDEMIOLOGY, CHAPTER 33

The Use of Pharmacoepidemiology to Study Beneficial Drug Effects

A Plain-Language Guide

Why proving a drug is safe and effective before approval is not the end of the story, and how researchers study whether it truly helps once it reaches everyday patients.

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What This Chapter Covers

To be approved for marketing, a drug must be proven safe and effective through premarketing trials. But those trials often cannot answer some of the most clinically important questions about a drug's benefits.

This chapter clarifies the different types of beneficial drug effects, explains why postmarketing studies of effectiveness are needed, and works through the methodologic problems — and solutions — involved in studying them.

Four Types of Drug Effects

Unanticipated Harmful

Unwanted effects no one could have predicted from preclinical or premarketing data. Chloramphenicol's link to aplastic anemia was not known at marketing.

Anticipated Harmful

Unwanted effects predicted ahead of time. Prazosin's first-dose fainting was known at marketing; the open question was how often it happened.

Unanticipated Beneficial

Desirable effects no one expected. Aspirin's power to help prevent a second heart attack was confirmed only after decades of use as a painkiller.

Anticipated Beneficial

The desirable effects a drug is prescribed for in the first place — the reason for the prescription, and the focus of this chapter.

The Aspirin Surprise

Aspirin's ability to lower the risk of a second heart attack was an unanticipated beneficial effect — confirmed only after the drug had already been used as a pain reliever for many years.

Because other nonaspirin anti-inflammatory drugs work through a similar mechanism, it seemed plausible they might share the same benefit. More recent data suggest the opposite instead: some of these medications may actually increase the risk of cardiovascular events.

FOUNDATIONS, 4 OF 4

Efficacy, Effectiveness, and Efficiency

Efficacy

Can the drug work under ideal conditions — perfect adherence, no interactions? Usually studied with a randomized trial.

Effectiveness

Does the drug work in the real world? An efficacious drug can still lack effectiveness, often because side effects undermine adherence. Effectiveness is generally best studied nonexperimentally after marketing, though pragmatic randomized trials can be used too.

Efficiency

Does the drug achieve its effect at an acceptable cost? This question belongs to health economics.

Each of these can be measured with clinical outcomes, patient-reported outcomes, utility measures, or economic outcomes — the same methodologic issues apply to all of them.

WHY POSTMARKETING STUDY MATTERS, 1 OF 7

The Clinical Questions That Matter

Before prescribing, a clinician really needs to know five things about a drug's intended benefit — and premarketing trials rarely answer all of them.

Can it have the desired effect at all? Does it actually achieve that effect in everyday practice? Does it have other beneficial effects, including long-term ones? Does it work better than the available alternatives? And how much do real-world factors change the size of the effect?

WHY POSTMARKETING STUDY MATTERS, 2 OF 7

Premarketing Trials See an Atypical World

Premarketing randomized trials generally show whether a drug can produce at least one beneficial effect, usually compared with a placebo.

But these studies are typically conducted in unusually controlled settings compared to where the drug will ultimately be used: adherence runs higher than in real practice, and participants tend to be similar in age and sex, free of other diseases, and not taking other interacting drugs.

The 100 Drug Uses Study

To measure the size of this gap, researchers compared the 100 most common drug uses in 1978 against what the FDA actually knew when it approved them.

Thirty-one of the 100 uses had not been FDA-approved at initial marketing, and eighteen still weren't approved years later. Eight of those eighteen were probably medically inappropriate, like using antibiotics for viral infections — but others could well have been entirely reasonable uses the regulatory process simply hadn't caught up with. Looking beyond those 100 uses at 52 related indications, researchers also found 8 where the drug's effects had never been tested against the available alternatives, leaving prescribers to choose among options with no head-to-head evidence at all.

WHY POSTMARKETING STUDY MATTERS, 4 OF 7

When Only Part of the Story Was Tested

Eight of the 100 common drug uses were based on an assumed long-term benefit, when only a shorter-term, intermediate effect had actually been studied before marketing.

Antihypertensive drugs are a clear example: they're used to prevent long-term cardiovascular complications, but were approved mainly on their proven ability to lower blood pressure. Hypoglycemic agents show the same pattern, approved for symptom control while their promise of preventing diabetes's vascular complications went untested.

Factors That Modify Drug Effects

Real-world factors can quietly change how well a drug works, and most were never fully explored before marketing.

Regimen

Furosemide for heart failure works better with more frequent, smaller doses than were first approved.

The Indication

Metaproterenol for asthma doesn't help severely ill patients without additional supplementary therapy.

The Patient

Diazepam's metabolism varies enormously with age, from premature infants to older children to the elderly.

Other Drugs and Diet

Diuretics can impair lithium clearance, and cutting sodium intake can improve a diuretic's own effect on blood pressure.

WHY POSTMARKETING STUDY MATTERS, 6 OF 7

The Hidden Complexity of Real Patients

Real patients rarely resemble the tidy trial population. They take other drugs, live with other conditions, and include children and pregnant women whom trials routinely exclude.

Additional prescriptions accompanied 62% of the prescriptions studied, and 41% were written for patients who had other illnesses besides the one being treated — yet the typical number of concomitant drugs and diagnoses was rarely fully explored before marketing.

Pediatric use ranged from 0% to 97% across the 100 drug uses studied, many of them essentially untested in children. And despite common use during pregnancy even then, only three of the drugs had actually been approved for pregnant patients.

Regulation Is Not Clinical Guidance

The FDA regulates drug manufacturers, not physicians. It doesn't tell doctors how to use a drug, and it doesn't initiate its own studies — it only evaluates what manufacturers submit.

The data needed to approve a drug and the data needed to guide a clinical decision are simply not the same thing. Given the regulatory focus, the atypical trial settings, and the practical limits of time, cost, and logistics on premarketing research, these gaps are not surprising — they are exactly why more postmarketing study of beneficial effects is needed.

THE CONFOUNDING PROBLEM, 1 OF 4

Confounding by Indication

When we study whether a drug produces its intended benefit, a distinctive problem arises: the very reason a patient received the drug can also predict how they fare.

The indication for treatment is a risk factor in its own right. It shapes who receives the drug and independently influences their outcome — so simply comparing treated and untreated patients can mislead us about the drug's real effect, in either direction.

The Beta-Blocker Example

Consider a beta-blocker prescribed after a heart attack to help prevent a second one.

Comparing treated and untreated patients might actually show a higher rate of recurrent heart attack among the treated group — not because the drug is harmful, but because the conditions that lead doctors to prescribe a beta-blocker, such as angina, arrhythmias, or hypertension, independently raise the risk of a second heart attack. A real benefit can be completely masked by the very reason the drug was given.

THE CONFOUNDING PROBLEM, 3 OF 4

When Confounding by Indication Isn't a Problem

This kind of confounding is generally not an issue when studying unexpected side effects that have little to do with why the drug was prescribed.

NSAIDs and gastrointestinal bleeding are a good example: whether someone takes the drug for arthritis, painful periods, or acute pain has little bearing on bleeding risk. But even here it can creep in — NSAID hypersensitivity reactions turned out to differ between acute-pain and chronic-pain use, likely because one involves intermittent dosing and the other continuous use.

THE CONFOUNDING PROBLEM, 4 OF 4

Can Only Trials Answer These Questions?

Some argue that confounding by indication is so fundamental that only a randomized trial can be trusted to study a drug's benefits.

But postmarketing trials carry many of the same problems as premarketing ones: they are logistically demanding, can raise ethical concerns once a drug is already in use, and still unfold in comparatively artificial settings. Rejecting nonexperimental research outright would mean giving up on answering many real-world questions at all. What's needed instead is a way to know which questions it can actually answer.

A Roadmap for Research Questions

Not every question about a drug's beneficial effects is equally vulnerable to confounding by indication. A four-category framework sorts questions by how serious the problem really is.

Some questions need no comparison group at all. Some have no confounding to worry about. Some have confounding that can be measured and controlled. And some have confounding that simply cannot be controlled with today's methods.

When No Comparison Is Needed

No Comparison Needed

Naloxone waking a patient from a methadone overdose, then the patient relapsing into coma as it wears off — that pattern needs no control group. The same is true for penicillin curing pneumococcal pneumonia across a series of patients.

Confounding Nonexistent

Giving a measles vaccine to healthy infants is a good example: the decision isn't based on differences between patients, so there's no indication-related confounding to control for in the first place.

CURRENTLY AVAILABLE SOLUTIONS, 3 OF 7

The Regression-to-the-Mean Trap

Even when a drug's effect looks obvious, one trap can still fool us.

Patients with recurring symptoms, like headaches, tend to seek treatment right when those symptoms are at their worst — and symptoms at their worst tend to improve on their own afterward, regardless of treatment. A drug given at that moment can get credit for an improvement that was going to happen anyway. Ruling out this possibility is essential before accepting an obvious-looking effect at face value.

CURRENTLY AVAILABLE SOLUTIONS, 4 OF 7

When Indication Can Be Measured and Controlled

The largest category covers questions where confounding by indication genuinely exists, but the indication itself can be measured well enough to control for it.

This works when the indication is a simple yes-or-no matter with no real gradations, when any gradations don't affect the treatment choice or the expected outcome, or when the indication can be thoroughly characterized and tracked even as treatment continues. In these situations, traditional tools — exclusion, matching, stratification, and mathematical modeling — can do the job.

Propensity Scores

A propensity score takes a different approach: instead of predicting the outcome, it predicts the exposure — the probability that a given patient would receive the drug at all.

Patients are grouped into categories of similar predicted probability, and those categories are used to control for confounding. The catch: this only works as well as the underlying data, since it still depends on actually measuring the true predictors of the treatment decision. Propensity scores tend to help most when outcome events are scarce relative to the number of confounders; with more events, conventional regression usually works just as well.

Instrumental Variables

An instrumental variable is a factor that's causally related to whether a patient gets the drug, only weakly related to any unmeasured risk factors, and not itself a step toward the outcome.

By controlling for the instrument instead of the indication directly, this method can address confounding that was never measured at all — a genuine advantage. The difficulty is finding one: a truly valid instrument is extremely hard to come by in pharmacoepidemiology, which limits how often this approach can actually be used.

When Confounding Cannot Be Controlled

In the last category, the indication creates confounding that current methods simply cannot measure or control.

Nonexperimental research can still offer qualitative, if not quantitative, support — but only when three conditions hold: the confounding must work in the opposite direction from the expected drug effect, that direction must be entirely predictable, and it must be large enough to rule out regression to the mean. Corticosteroids for status asthmaticus fit this pattern: steroid-treated patients are expected to be sicker, so if they actually recover faster, that argues for a real benefit despite the confounding. This logic works in only one direction: a better outcome despite predicted disadvantage supports benefit; failure to see a better outcome does not establish lack of benefit.

HOW OFTEN DOES THIS WORK, 1 OF 2

Putting the Framework to the Test

How often can this framework actually deliver an answer? Researchers tested it against the 100 most recently approved new drugs as of December 1978.

Excluding seven contact-lens-related products, 93 drugs were examined across all of their indications — 131 separate drug-indication pairs in total, each one classified using the four-category framework.

HOW OFTEN DOES THIS WORK, 2 OF 2

What the Study Found

Evaluating the Drug for Its Indication

89 of 131

Simple clinical observation sufficient, no comparison needed.

42 of 131

Formal comparative research required:

7 — confounding not an obstacle

35 — confounding uncontrollable

Of those 42 requiring comparative research, 30 could not ethically use a placebo-controlled trial.

Comparing One Active Drug With Another

All 131 required formal comparative research. 94 of 131 (71.8%) were theoretically suitable for valid nonexperimental study, and an experimental active-comparator trial was ethically possible for all 131.

“Theoretically studiable” does not guarantee a valid real-world study.

SPECIFIC EXAMPLES, 1 OF 11

Estrogens and Fractures

Could estrogen therapy prevent osteoporotic fractures? The first in a series of real examples showing how researchers have navigated confounding by indication in practice.

Biochemical studies and bone-density data suggested estrogens might reduce fracture risk — but no study had directly examined fractures themselves.

SPECIFIC EXAMPLES, 2 OF 11

Why a Trial Wasn't Practical

A randomized trial faced two serious obstacles.

Fractures are too uncommon within any workable time period to make the necessary sample size realistic, and a trial would need years of follow-up to detect a difference. Nonexperimental case-control and cohort studies were used instead — but most didn't examine why some women received estrogens and others didn't, leaving room for confounding by indication.

What the Estrogen Studies Found

Three studies documented important differences between treated and untreated women — in weight, age at menopause, alcohol use, height, blood pressure, marital status, race, and economic status, among other factors.

In the detailed adjusted analysis described by the chapter, the protective association remained — but the reason some women received estrogen and others did not was still unknown. The finding was later confirmed by a randomized trial: the Women's Health Initiative.

SPECIFIC EXAMPLES, 4 OF 11

Anticoagulants and Recurring Blood Clots

Both intravenous and oral anticoagulants clearly reduce the recurrence of venous thromboembolism. The open question was how long treatment should continue.

For decades, the evidence on optimal duration was mostly anecdotal, drawn from case series at a single center. The one long-standing randomized trial — six weeks versus six months of treatment, 186 subjects, only 7 recurrences, more than half with a transient risk factor — found no difference between the two durations.

SPECIFIC EXAMPLES, 5 OF 11

The Kaiser Anticoagulation Study

A large retrospective cohort study at Kaiser Permanente took a different approach: ten years of data, 1.6 million people, 16 million patient-years, 3,384 hospitalized cases of venous thromboembolism, 2,473 of them idiopathic.

It found that prolonging treatment beyond six weeks sharply increased the risk of major bleeding, with no benefit for preventing recurrence. The key to trusting this result: physicians' choice of treatment duration was essentially random and spread evenly across the follow-up period, so indication wasn't confounding the comparison.

SPECIFIC EXAMPLES, 6 OF 11

The Swedish Trial Complicates the Picture

A decade later, a multicenter Swedish randomized trial of 897 patients with a first venous thromboembolism told a different story.

Over two years of follow-up, recurrence was 18.1% in the six-week group versus 9.5% in the six-month group — a significant difference, with no difference in mortality or major bleeding. Several other studies have since shown a similar benefit from longer warfarin treatment, contradicting the Kaiser finding. Even a well-designed nonexperimental study can be overturned by later evidence.

SPECIFIC EXAMPLES, 7 OF 11

Lidocaine and Heart Attacks

A case-control study, matching heart attack deaths against survivors by age, gender, race, and hospitalization date, looked at whether lidocaine could prevent death after a heart attack.

Lidocaine showed no protection against death overall, and helped only when deaths from ventricular arrhythmia were analyzed on their own. The investigators tried to control for underlying risk by classifying heart failure status, but that classification was too simple to capture true risk — and the result matched a randomized trial from around the same time. More than twenty later trials and four meta-analyses indicated that lidocaine reduces ventricular fibrillation but increases overall mortality after a heart attack, the opposite of what the earlier nonrandomized studies suggested — although a later reanalysis of 43,704 patients from the GUSTO-I and GUSTO-IIb trials did not confirm the mortality finding.

SPECIFIC EXAMPLES, 8 OF 11

Anticoagulants and Heart Attack Deaths

Could anticoagulants prevent death after a heart attack? Randomized trials gave inconsistent, inconclusive results, possibly because of small sample sizes.

A case-control study tried to control for underlying risk the same way the lidocaine study did, and ran into the same doubts about whether physicians' risk judgments can really be measured accurately. A more convincing design might have compared doctors who always prescribed anticoagulants against those who never did — though even this natural experiment can fail if prescribers who favor one strategy differ in other ways that affect their patients' outcomes. In the end, this was a question that really needed randomized trials, and many have since been done as newer anticoagulant and clot-dissolving therapies became available.

Generic Versus Brand-Name Drugs

Since 1984, generic drugs have been allowed to market after demonstrating bioequivalence in a small study of 18 to 24 healthy adults — but bioequivalence doesn't guarantee clinical equivalence.

If a generic and brand-name drug do differ clinically, that difference is more likely to show up in beneficial effects than in toxicity. Doses are optimized for efficacy, while toxicity typically occurs at higher concentrations — so a modest variation in plasma levels is more likely to affect how well a drug works than how safe it is. Even a change in a pill's appearance can reduce adherence.

SPECIFIC EXAMPLES, 10 OF 11

The COMPASS Database Studies

Studying generic equivalence nonexperimentally makes sense: detecting a small difference needs an enormous sample size, and since a physician is comparing versions of the same drug, the choice usually comes down to pharmacy stock rather than the patient's condition.

Unpublished studies using the COMPASS database compared patients who switched from brand to generic against those who didn't, for thioridazine, chlorpropamide, and a slow-absorption theophylline. Even before confounding, the comparison faced data problems: researchers often couldn't confirm which manufacturer's product a patient actually received, and had to rely on proxy measures of efficacy, such as physician visits, hospitalizations, and use of add-on therapy. A baseline check, done before any switching occurred, then found that future switchers already differed from future non-switchers in all three studies. Confounding by indication was operating after all — so the planned efficacy comparisons were never carried out, and the results were never published.

Cost-Effectiveness Studies

Cost-effectiveness research asks how many resources are needed to achieve a beneficial outcome, combining a clinical measure with an economic one.

Confounding by indication can distort both halves of that equation, and not necessarily in the same way. A sicker population might cost more to treat regardless of the drug's effect, while also appearing to respond differently. Any cost-effectiveness study needs to think through the indication's implications for the clinical outcome and the cost variable separately.

Vaccines

Nonexperimental case-control and cohort studies have been widely used to study vaccines — for pneumococcal, rubella, measles, Haemophilus influenzae type b, oral poliovirus, meningococcus, Japanese encephalitis, BCG, diphtheria toxoid, mumps, leprosy, and pertussis vaccines.

Randomized trials are difficult here because disease is often too infrequent in a partly vaccinated population — though not impossible. Vaccinated and unvaccinated people can still differ in socioeconomic status, healthcare access, and physicians' attitudes toward the vaccine. But what makes indication confounding relatively manageable for vaccines is more specific: doctors usually don't select among otherwise eligible patients based on their perceived risk of actually contracting the disease. This has become a standard, accepted approach.

Other Case-Control Examples

Nonexperimental designs have answered a range of other beneficial-effect questions along similar lines.

Bicycle Safety Helmets

Studied for their effect on preventing facial injury.

Antibiotic Prophylaxis

Studied for preventing endocarditis following dental procedures.

Beta-Blockers

Studied for preventing death after a heart attack, and preventing a first coronary event.

The RCT-Only Debate

Clinicians have long valued careful clinical observation, and much of what we know about drugs' usefulness today comes from nonexperimental research — though it can't be accepted uncritically.

Some influential voices have argued that the randomized trial is the only reliable way to know whether a treatment works, going so far as to urge clinicians to discard at once all articles on therapy that are not randomized trials. This chapter argues that view is too simplistic and too far-reaching — it risks leaving necessary, available information uncollected and unused.

Bradford Hill's Warning

Sir Austin Bradford Hill, the very statistician who developed the modern randomized controlled trial, offered a caution of his own.

“Any belief that the controlled trial is the only way to study therapeutic efficacy would mean not only that the pendulum had swung too far but that it had come right off its hook.”

— Sir Austin Bradford Hill

Real-World Evidence Today

Many investigators now apply nonexperimental designs to studying beneficial effects, including head-to-head comparisons of active treatments — the same territory covered by comparative effectiveness research.

The 21st Century Cures Act saw the US Congress formally embrace real-world evidence as a source of information about drug effects, including benefit. Confounding by indication still demands careful attention, though more tools are becoming available. Nonexperimental research is unlikely to be as powerful as a well-run trial, and it is not offered as a replacement for one — but when a trial is unnecessary, unethical, infeasible, or too costly relative to what it would teach us, it remains a reasonable and valuable alternative.

Key Takeaways

Knowing that a drug works isn't the same as knowing it will help the patient in front of you. Premarketing trials answer a narrower question, in a narrower setting, than most clinical decisions require.

Confounding by indication is the central challenge in studying a drug's intended benefits — but it doesn't rule out nonexperimental research. It rules out treating every question the same way.

A trial is not always necessary, ethical, or feasible. When it isn't, a careful nonexperimental study, matched to the right category of question, can still yield a trustworthy answer.

WRAPUP, 2 OF 2

Thank You

A drug's intended benefit is often the hardest of its effects to pin down — and pharmacoepidemiology's job is to pin it down anyway, one carefully matched question and method at a time.

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